

Why Average Price Alone Is Too Narrow for Market Monitoring

A data analytics case study in market monitoring and KPI design for the Spanish day-ahead electricity market (2018–2026)

Kai A. Sanchez P.

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Executive Summary

This project asks a practical monitoring question in the Spanish day-ahead electricity market: is average daily wholesale price enough to summarize market stress?

The answer is no. Average daily price remains a useful headline KPI, but it compresses the day into one number and can miss operationally relevant information. Two days with similar average prices can still differ materially in intraday dispersion, concentration of high-price hours, and the broader shape of stress across the day.

To address that limitation, the project builds and evaluates a broader monitoring KPI. The final framework distinguishes between a core stress score based on price level, intraday range, and high-price-hour concentration, and an extended score that adds cleared-curve complexity as a secondary interpretive signal.

The main result is deliberately nuanced. Under an internal recall-style benchmark, the composite score does not outperform simpler one-metric rules such as average daily price or daily range. That rules out the strongest possible claim.

The score's value lies elsewhere: it changes which days rise to the top of the review queue and helps identify a distinct, interpretable class of multidimensional stress days that single-metric rules do not prioritize in the same way.

This is therefore a KPI-design and monitoring case study, not a structural model of electricity prices. Its contribution is practical: a one-number dashboard is too narrow for operational monitoring, and a broader descriptive lens can improve triage, review, and communication.

Business Problem

A market dashboard often leads with one number: average daily price. That KPI is useful for quick comparison across days, months, and years. But it can also hide information that matters operationally.

For a market monitoring analyst, that creates a practical problem:

- a day with a moderate average price can still have large within-day volatility,
- stress can be concentrated in a subset of hours rather than spread evenly,

- and some days deserve analyst attention not because they are simply expensive, but because stress is broad across multiple dimensions.

In other words, a price-only dashboard can mis-rank review effort. Some expensive days are operationally straightforward, while some moderate-price days deserve review because intraday stress is wider or more irregular than the headline average suggests.

Data and Pipeline

The project combines two (Nominated Electricity Market Operator) OMIE-based inputs:

1. Spanish day-ahead hourly prices
2. Day-ahead cleared curves, summarized into hourly market features

The main data-engineering challenge is that OMIE curve files change structure over time. The pipeline standardizes multiple raw formats into a single clean panel, summarizes curves to hourly features, merges those features with hourly prices, and aggregates the merged panel to daily monitoring features.

The final daily analytical panel uses the intersection of hourly prices and hourly cleared-curve summaries. This prioritizes comparability and integrity over maximal raw coverage.

Start date	End date	Daily observations	Average daily price (EUR/MWh)	Average daily range (EUR/MWh)
2018-01-01	2026-03-09	2986	78.5	50.8

Table 1: Analytical panel coverage.

Reproducibility

The project is designed as a script-based analytical pipeline rather than a single notebook workflow. Raw OMIE files are downloaded, standardized, summarized, merged, and validated through sequential R scripts, with intermediate outputs written to `data_clean/` and final tables and figures written to `outputs/tables/` and `outputs/figures/`.

In practical terms, the report depends on the following stages:

1. price panel construction,
2. raw curve parsing and standardization across file formats,
3. hourly curve summarization,
4. hourly merge with prices,
5. daily feature construction,
6. exploratory analysis, score construction, validation, and QA table generation.

This structure improves reproducibility in two ways. First, each stage leaves auditable intermediate files rather than only in-memory objects. Second, the final report reads pre-built analytical outputs, which separates data engineering, KPI construction, and reporting into distinct layers.

To reproduce the final outputs, the scripts should be run in pipeline order before knitting this report. The report then loads the cleaned daily panels, validation outputs, and generated figures/tables directly from disk.

The main execution order is: price panel construction, curve parsing, hourly curve summarization, hourly merge, exploratory analysis, score validation, and QA summary generation, assuming raw downloads already exist.

Data Quality and QA Checks

Before constructing the analytical panel, I ran integrity checks on the raw OMIE curve files, the summarized hourly curve panel, and the merged hourly price-curve panel. These checks focused on format consistency, parsing losses, duplicate date-hour rows, retained overlap, and DST handling.

metric	value
Raw curve formats detected	3 formats: legacy hourly, numeric Periodo hourly, quarter-hourly Periodo
Raw curve files processed	2991
Curve files flagged as defective	0
Rows kept after curve parsing filters	324394781
Dropped curve row share	0.000114
Share of days with quarter-hourly source rows	0.053828
Days with quarter-hourly source rows	161
Average quarter-row share within hourly summaries	0.053834
Raw price files processed	2986
Rows kept after price parsing filters	71656
Hourly merge duplicate date-hour rows	0
Hourly rows in merged panel	71656
Days in merged panel	2986
Daily coverage retained vs prices	1
Daily coverage retained vs curves	0.998328

Table 2: Summary of key data-quality and integrity checks.

These checks suggest that the analytical panel is not being driven by duplicate joins, DST mishandling, or major parsing failures. The merged panel retains full daily coverage relative to prices and nearly full daily coverage relative to curves, with expected 23-hour DST days preserved.

Finding 1: Average Daily Price Is Useful, but Incomplete

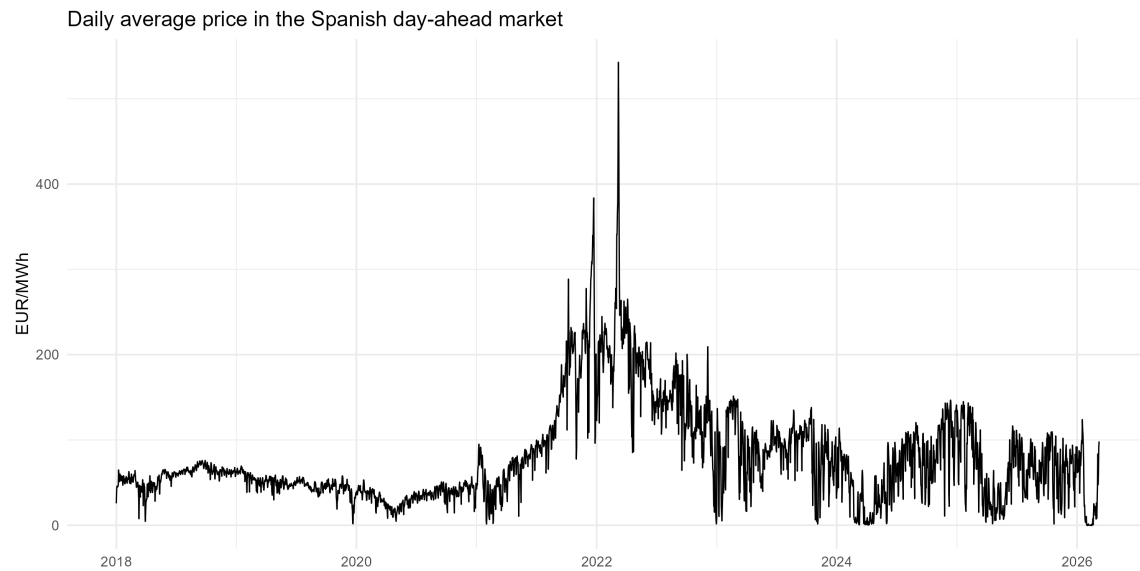


Figure 1: Daily average price in the Spanish day-ahead market.

Figure 1 provides the standard market view. It captures broad regime changes clearly, including the 2021–2022 stress episode and the lower-price environment that follows.

As a headline KPI, this figure is useful. It summarizes the market in a compact and intuitive way. But that is also its limitation: a daily average cannot tell us whether the day was relatively smooth or whether stress was concentrated in a handful of hours.

Finding 2: Intraday Range Reveals Hidden Stress

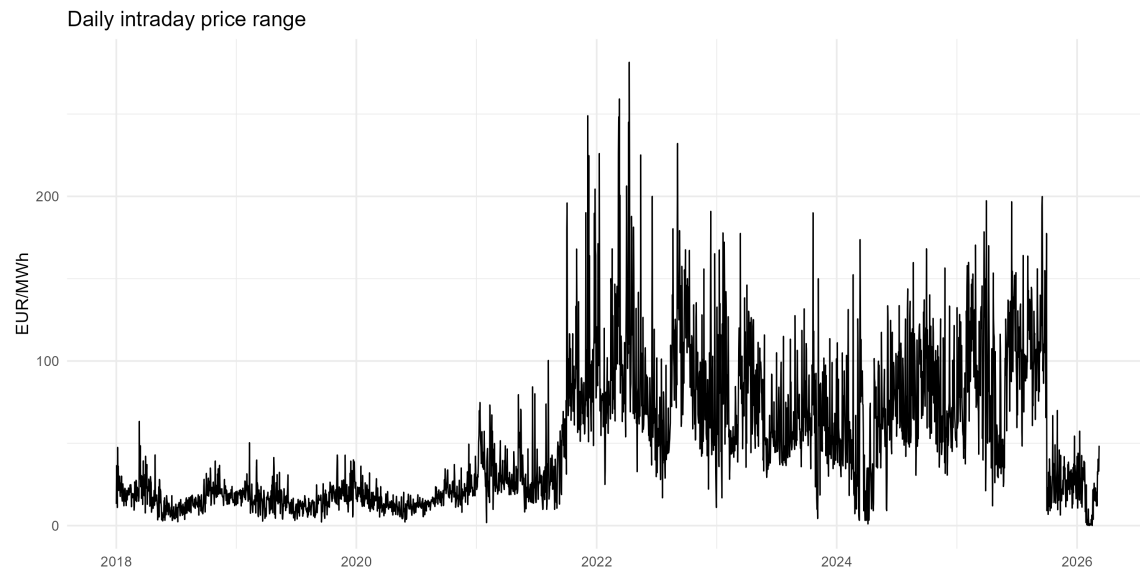


Figure 2: Daily intraday price range.

Figure 2 shows the daily intraday range: the difference between the highest and lowest hourly prices within a day.

This is where the weakness of the average becomes visible. There are periods where average daily price is not especially extreme, but within-day range is still large. A dashboard that only reports average daily price would miss that dimension.

Finding 3: Similar Average-Price Days Can Behave Very Differently

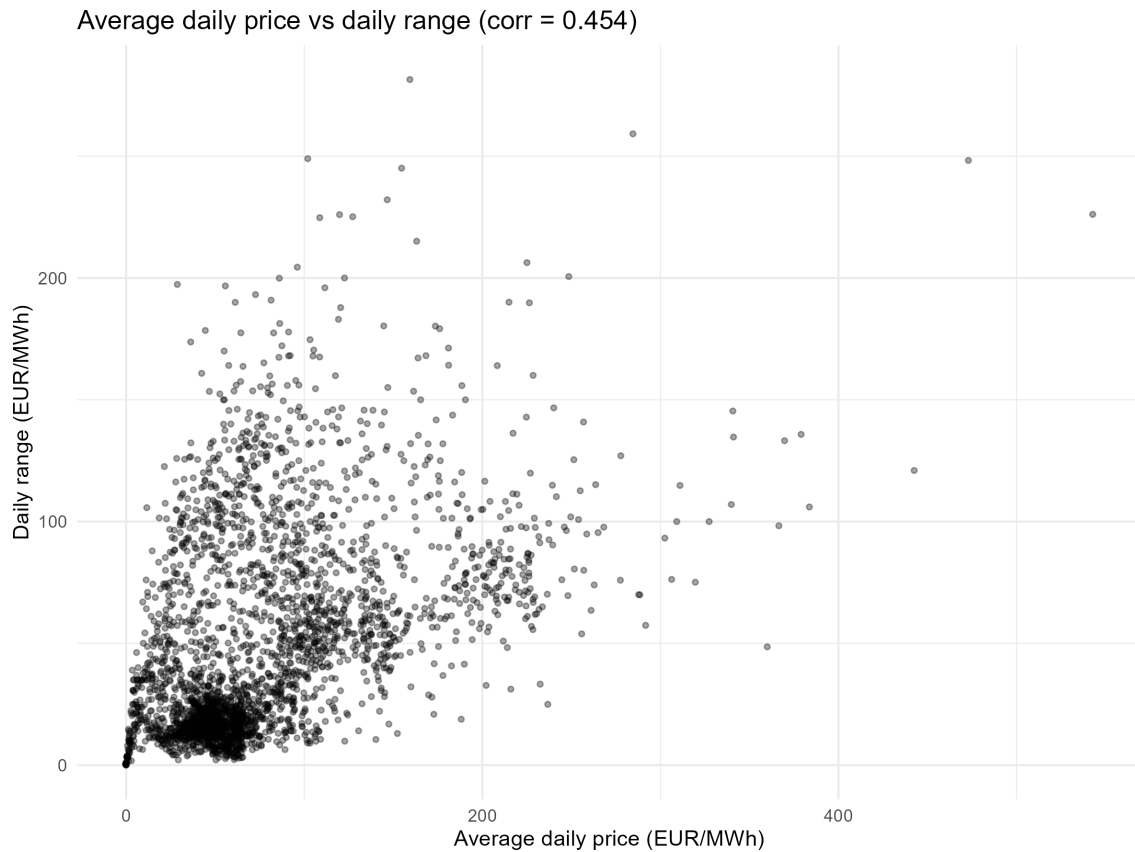


Figure 3: Average daily price versus daily intraday range.

Figure 3 compares average daily price and daily intraday range directly.

There is a positive relationship, but it is far from one-to-one. The correlation between the two metrics is 0.454. Within similar average-price ranges, daily range still varies substantially. That is the operational gap a price-only KPI leaves unresolved.

Finding 4: A Broader Monitoring Score Improves the Lens

The project translates that critique into a practical monitoring KPI.

The core stress score combines:

- 40% daily average price
- 40% daily intraday range
- 20% number of high-price hours

The extended score adds:

- 10% cleared-curve complexity

Before weighting, each score component is winsorized and rescaled to the unit interval. This ensures that the score combines relative position across dimensions rather than raw values in different units. The weighting scheme is intentionally heuristic and monitoring-oriented. It is designed for interpretability and operational triage rather than statistical optimization.

This distinction matters. The core score uses the strongest and most stable descriptive signals. The extended score preserves some structural information from the cleared curves, but treats it as a secondary interpretive signal rather than a dominant driver.

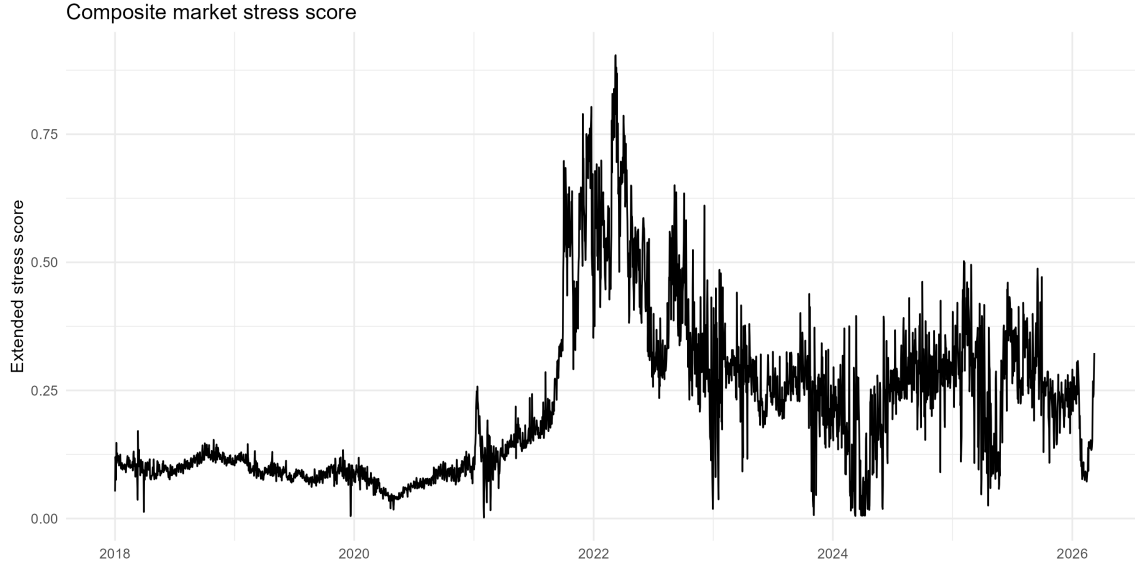


Figure 4: Composite market stress score (extended version).

Variant	Weights	Correlation with base	Top-20 overlap with base
base	35/35/20/10	1.0000000	1.0
core	40/40/20/0	0.9916114	1.0
no_steps	40/40/20/0	0.9916114	1.0
price_range_only	50/50/0/0	0.9749569	0.8

Table 3: Sensitivity of the score to alternative weighting schemes.

This helps answer a basic robustness question: if the weights move moderately, does the ranking collapse? In this project, the answer appears to be no. That does not prove the weights, but it shows that the ranking is not purely an artifact of one exact specification.

Score Validation: Useful, but Not Universally Superior

This benchmark is deliberately narrow and internal. It tests how well each metric recovers internally defined one-dimensional stress events, not whether a metric is operationally superior under external ground truth.

Internal stress events are defined as days in the top 5% of at least one of three variables: average daily price, daily range, or number of high-price hours.

Because the internal event definition is constructed from variables that are closely related to the score and its comparators, this benchmark should be interpreted cautiously. It is useful for comparing internal alignment across candidate KPIs, but it is not an external validation of operational superiority. In that sense, the exercise measures relative consistency with internally defined stress conditions more than real-world ground truth.

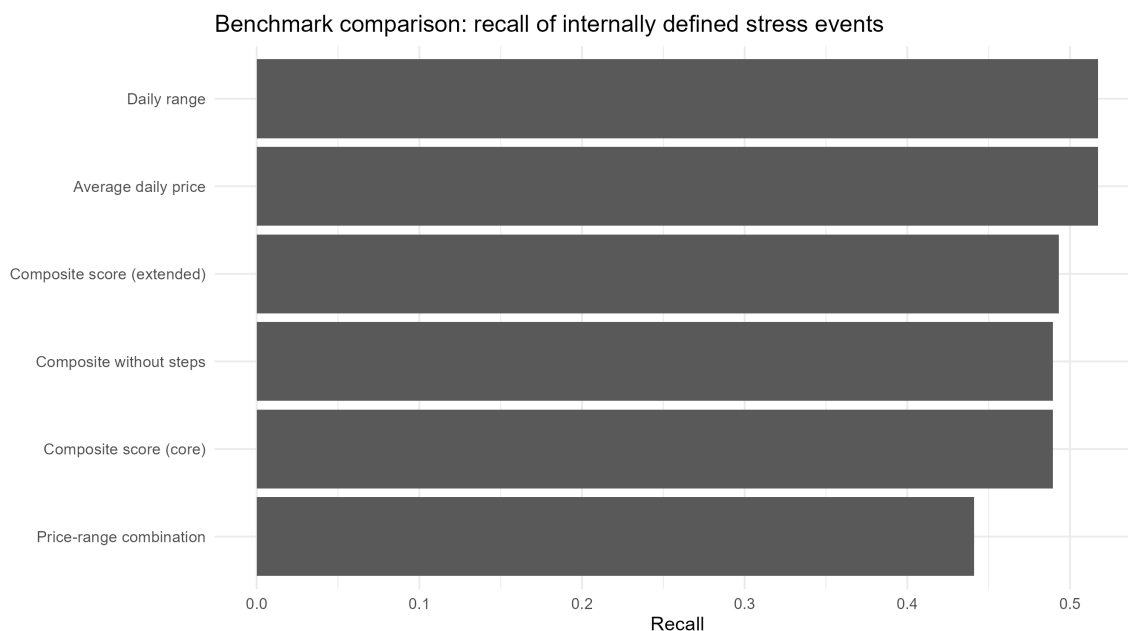


Figure 5: Benchmark comparison: recall of internally defined stress events.

Metric	Flagged days	Captured internal stress	Precision	Recall
Average daily price	149	149	1.000	0.517
Daily range	149	149	1.000	0.517
Composite score (extended)	149	142	0.953	0.493
Composite score (core)	149	141	0.946	0.490
Composite without steps	149	141	0.946	0.490
Price-range combination	149	127	0.852	0.441

Table 4: Internal validation: composite score versus simpler benchmarks.

The result is clear. On this internal recall-based benchmark, the composite score does not outperform average daily price or daily range. That rules out the strongest possible claim.

At the same time, the result should not be read as evidence that the score is uninformative. The core and extended versions still identify a concentrated subset of high-stress days with interpretable profiles. The practical implication is narrower and more useful: simple one-metric rules remain

strong for this internal recall task, while the composite score remains valuable as a broader prioritization and triage metric.

The correct interpretation is narrower and more credible:

The composite score is best understood as a broader monitoring and prioritization metric, not as a universally dominant benchmark.

Incremental Value: The Score Flags Different Kinds of Days

The most useful result from the benchmark exercise is not recall alone, but incremental value. The composite score and single-metric rules do not flag exactly the same days. That means the score is not simply reproducing price or range with more arithmetic; it changes which days move to the top of the review queue.

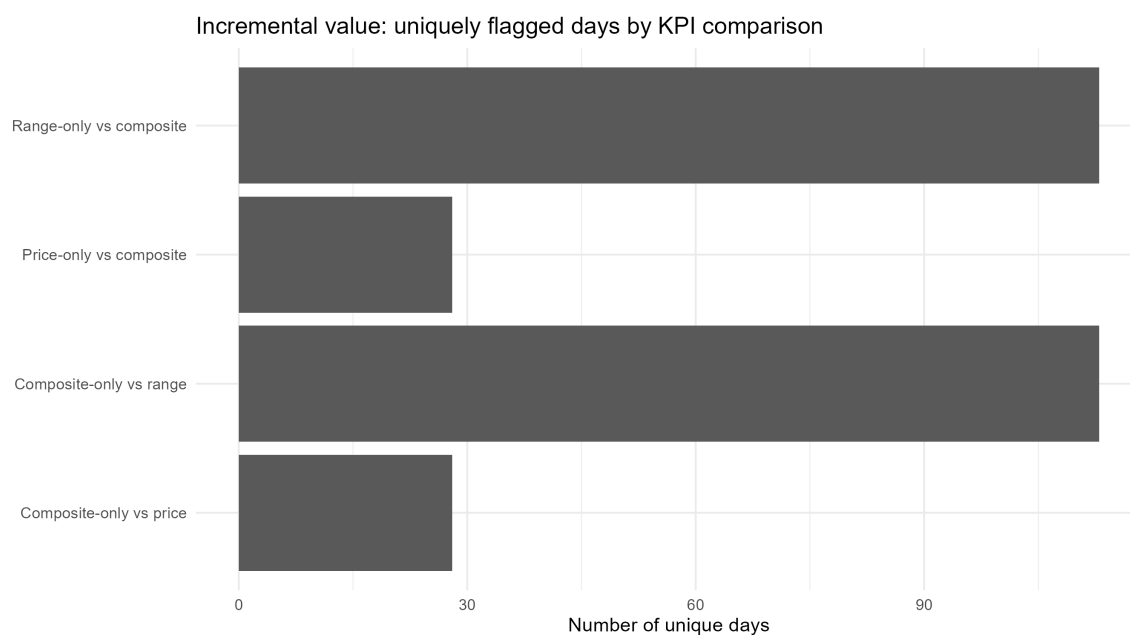


Figure 6: Incremental value: uniquely flagged days by KPI comparison.

Comparison	Number of unique days
Composite-only vs price	28
Price-only vs composite	28
Composite-only vs range	113
Range-only vs composite	113

Table 5: Uniquely flagged days across KPI comparisons.

Profile Differences Across Uniquely Flagged Days

Group	Days	Avg price	Avg range	Avg high-price hours	Avg score
Composite-only vs price	28	172.1	161.7	8.00	0.597
Price-only vs composite	28	207.4	70.5	11.82	0.520
Composite-only vs range	113	239.5	86.3	19.11	0.643
Range-only vs composite	113	87.1	154.3	0.48	0.427

Table 6: Average profiles of uniquely flagged day groups.

These groups differ not only in membership, but in character.

- Composite-only days relative to price tend to have lower average price, much wider daily ranges, and fewer high-price hours than price-only days.
- Range-only days relative to the composite tend to reflect sharper intraday instability without equally broad stress across the day.

That is the score’s main contribution. It does not maximize recall of narrow internal event rules. It re-ranks attention toward days where stress is broader than a single dimension.

Illustrative Examples

Date	Avg price	Daily range	High-price hours	Composite score
2021-10-02	173.6	180.2	13	0.698
2021-10-11	184.5	116.3	9	0.555
2021-10-20	201.4	116.5	9	0.580
2021-12-29	181.1	164.2	12	0.672
2022-01-05	181.0	171.2	11	0.678
2022-01-09	119.8	226.0	6	0.566

Table 7: Illustrative composite-only days relative to a price-only rule.

Date	Avg price	Daily range	High-price hours	Composite score
2021-10-01	216.0	31.2	20	0.521
2021-10-05	203.7	91.5	10	0.540
2021-10-17	209.6	86.2	9	0.529
2021-10-19	207.1	84.7	11	0.541
2021-10-21	207.9	69.4	11	0.512
2021-10-22	203.5	82.2	11	0.531

Table 8: Illustrative price-only days relative to the composite score.

Date	Avg price	Daily range	High-price hours	Composite score
2021-10-03	111.6	196.0	2	0.521
2021-10-30	124.5	133.3	0	0.432
2021-11-01	91.2	168.0	0	0.456
2021-11-07	132.5	136.0	0	0.449
2021-12-05	102.0	249.0	6	0.542
2021-12-08	108.7	224.8	4	0.534

Table 9: Illustrative range-only days relative to the composite score.

Supporting Evidence: Hourly Structure Still Matters

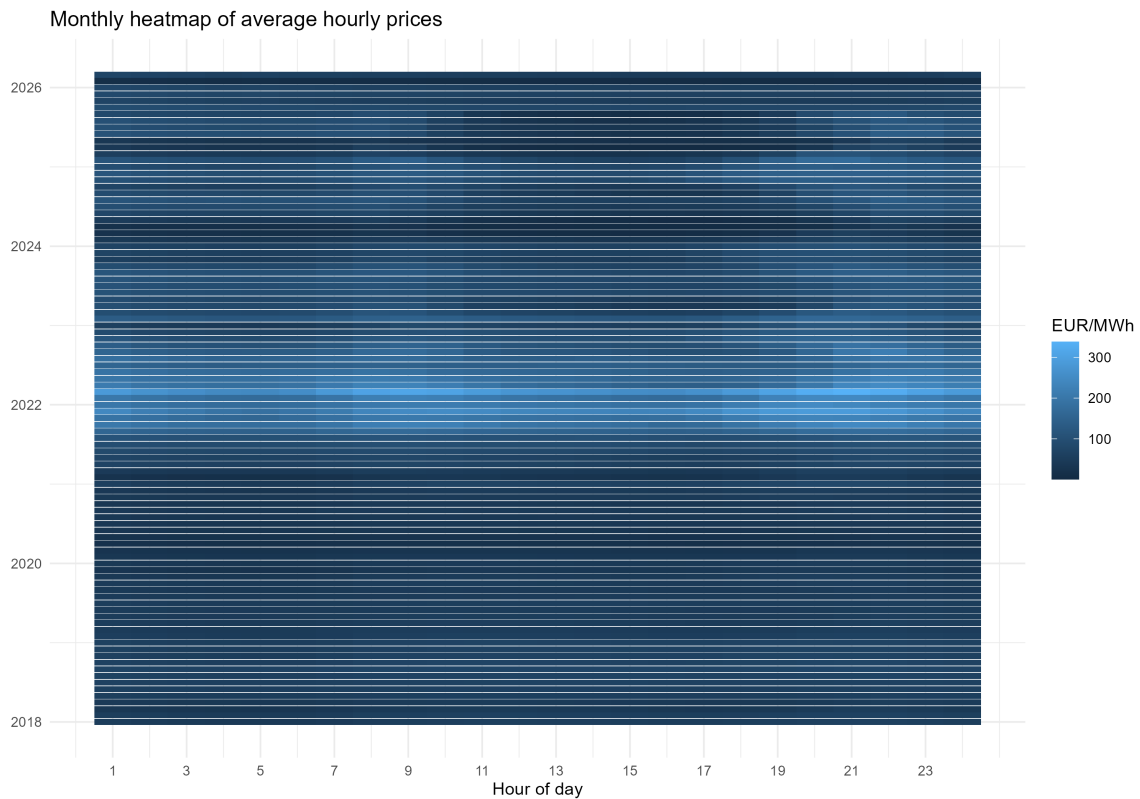


Figure 7: Monthly heatmap of average hourly prices.

The market has a strong and persistent hourly structure. Some hours are systematically more expensive than others, and some periods show much more pronounced within-day variation than the daily average alone would suggest.

When Would I Use Which KPI?

KPI	Best use	Weakness
Average daily price	Headline comparison across days and months	Hides intraday structure
Daily range	Detecting within-day instability	Can overemphasize narrow spikes
High-price hours	Capturing persistence of elevated prices	Depends on threshold choice
Composite score	Triage and prioritization of broad stress days	Judgment-based and not benchmark-dominant

Table 10: Best use and weakness of each KPI in a monitoring workflow.

No single metric does everything well. The headline average is good for comparison. Daily range is good for spotting instability. High-price hours help identify persistent expensive conditions. The composite score is most useful when the analyst wants to rank broad stress days rather than simply count expensive or high-dispersion episodes.

From Analysis to Monitoring Product

The project is most useful when translated into a monitoring workflow. A dashboard could show:

- average daily price,
- daily intraday range,
- high-price-hour concentration,
- composite stress score,
- and an alert bucket.

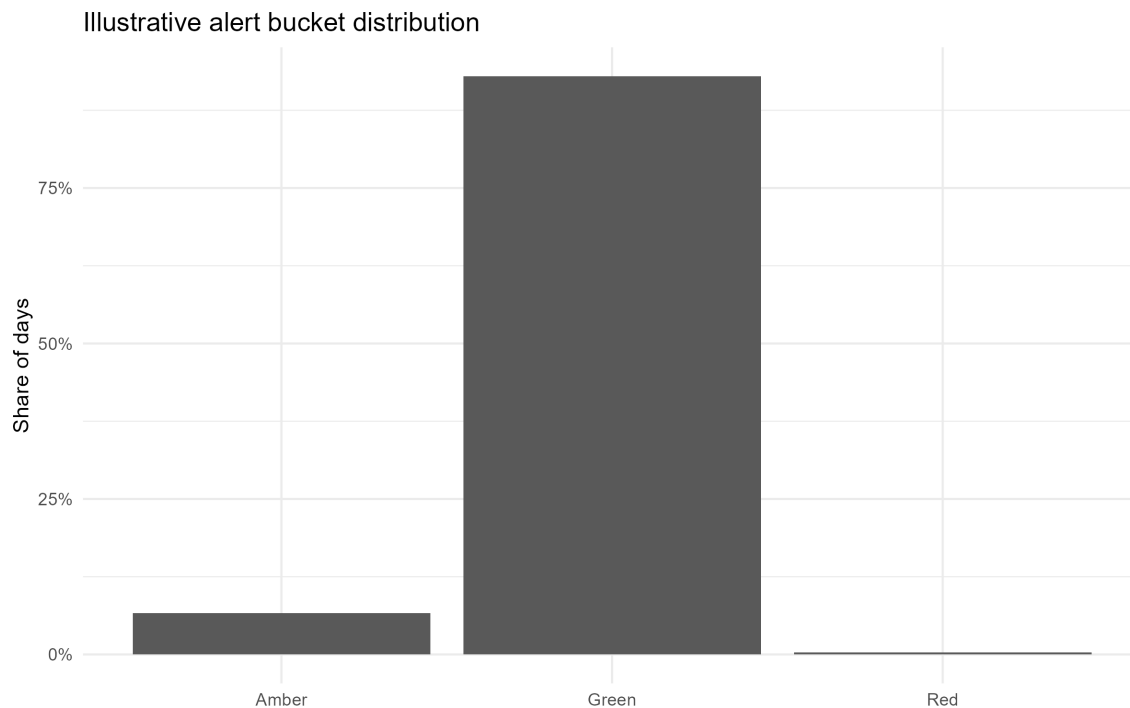


Figure 8: Illustrative alert bucket distribution.

Bucket	Rule	Interpretation
Green	$\text{score} < 0.50$	Normal monitoring conditions
Amber	$0.50 \leq \text{score} < 0.80$	Elevated stress worth monitoring
Red	$0.80 \leq \text{score} < 0.95$	High-priority review tier
Extreme	$\text{score} \geq 0.95$	Extreme stress escalation tier

Table 11: Illustrative alert thresholds for a monitoring workflow.

Bucket	Share of days
Green	93%
Amber	6.7%
Red	0.4%

Table 12: Observed distribution of alert buckets in the daily panel.

Condition	Interpretation	Suggested action
High score + high average price	Broad and persistent expensive conditions	Escalate for analyst review
High score + wide daily range + fewer high-price hours	Multidimensional stress without uniformly high prices	Review intraday structure and specific hours
Low score + high average price	Persistently expensive but less multidimensional day	Track as headline expensive day
High range-only profile	Sharp intraday dispersion rather than broad stress	Flag for volatility-focused review

Table 13: Illustrative decision rules for analyst review.

Even without formal calibration, the bucket framework helps translate a descriptive score into a more structured analyst workflow.

Limitations

This is a descriptive monitoring project, not a structural market model.

Several limitations define the scope of this project.

- The benchmark is internal and partially self-referential: stress events are defined using variables closely related to the score and its comparators. As a result, the validation tests internal alignment rather than external operational performance.
- The weighting scheme is transparent and interpretable, but still judgment-based rather than optimized against external outcomes.
- The high-price-hours component is threshold-dependent, which makes the score partly sensitive to the choice of cutoffs.
- The final score depends on prior winsorization and rescaling choices, which are necessary for comparability across components but not innocuous.
- Cleared-curve complexity is conceptually useful but less comparable across file-format and market-period regimes, especially toward the end of the sample.
- The project does not validate the score against external labels such as known market incidents, analyst review logs, or ex post operational classifications.

These limitations do not eliminate the score’s usefulness. They define its scope correctly.

Next-Step Validation

A stronger next version of the project would validate the score against semi-external or external outcomes, such as known market incidents, analyst review notes, balancing-market volatility, interconnection disruptions, fuel-price shocks, or manually labeled stress episodes. That would make it possible to test whether the composite score delivers a measurable operational gain relative to simpler rules—for example, earlier detection of relevant days, better top-N prioritization, or lower false-positive rates for certain types of review.

Conclusion

This project shows that average daily wholesale price is a useful but incomplete summary of market stress in the Spanish day-ahead electricity market.

Four conclusions follow:

1. Average daily price is necessary, but too narrow on its own.
2. Daily range and high-price concentration add meaningful monitoring information.
3. The composite score is not a universal winner on internal recall-based validation.
4. The composite score still adds value as a prioritization tool by identifying a distinct and interpretable class of multidimensional stress days.

For monitoring purposes, the practical lesson is not that headline price is useless, but that it is too narrow to rank market stress on its own.

Appendix

Appendix A: Supporting Tables

Annual summary

Year	Avg daily price	Avg daily range	Avg high-price hours	Avg cleared steps	Avg core score	Avg extended score	Days
2018	57.4	17.4	0.00	1244	0.120	0.106	365
2019	47.9	16.3	0.00	1267	0.102	0.091	365
2020	34.0	16.3	0.00	1380	0.081	0.074	365
2021	112.0	48.8	3.32	1452	0.300	0.270	365
2022	167.9	92.7	6.46	1578	0.509	0.458	365
2023	88.3	69.9	0.02	1677	0.288	0.259	364
2024	63.4	69.6	0.00	1748	0.248	0.226	366
2025	64.9	81.2	0.04	3518	0.277	0.274	363
2026	36.8	19.0	0.00	8164	0.093	0.174	68

Table 14: Annual summary statistics.

The visible increase in cleared steps in 2025–2026 should be interpreted with caution. Because cleared-step counts are less comparable across file-format and market-period regimes, this feature is treated in the main analysis as a secondary interpretive signal rather than a core ranking driver. That is precisely why the project distinguishes between a core score and an extended score.

Monthly summary (sample)

Year	Month	Avg daily price	Avg daily range	Avg total steps	Avg extended score
2018	Jan	51.6	22.2	1198	0.108
2018	Feb	55.0	19.3	1223	0.106
2018	Mar	39.8	25.5	1177	0.098
2018	Apr	42.7	20.8	1191	0.092
2018	May	55.1	12.4	1256	0.093
2018	Jun	58.5	9.1	1280	0.092
2018	Jul	61.8	10.0	1298	0.098
2018	Aug	64.3	13.6	1256	0.108
2018	Sep	71.3	14.0	1266	0.119
2018	Oct	65.4	22.6	1251	0.128
2018	Nov	62.0	22.2	1286	0.123
2018	Dec	61.9	17.3	1250	0.112
2019	Jan	62.7	18.9	1273	0.117
2019	Feb	54.7	18.4	1270	0.105
2019	Mar	49.2	15.3	1253	0.091
2019	Apr	50.7	16.6	1250	0.095
2019	May	48.7	16.7	1278	0.094
2019	Jun	47.2	12.1	1309	0.083
2019	Jul	51.5	11.8	1318	0.088
2019	Aug	45.0	10.3	1254	0.075
2019	Sep	42.1	14.3	1256	0.080
2019	Oct	47.2	17.7	1262	0.093
2019	Nov	42.1	22.0	1264	0.095
2019	Dec	33.7	22.1	1221	0.083

Table 15: Monthly summary (first 24 rows shown).

Appendix B: Supporting Figures

Hourly average price profile

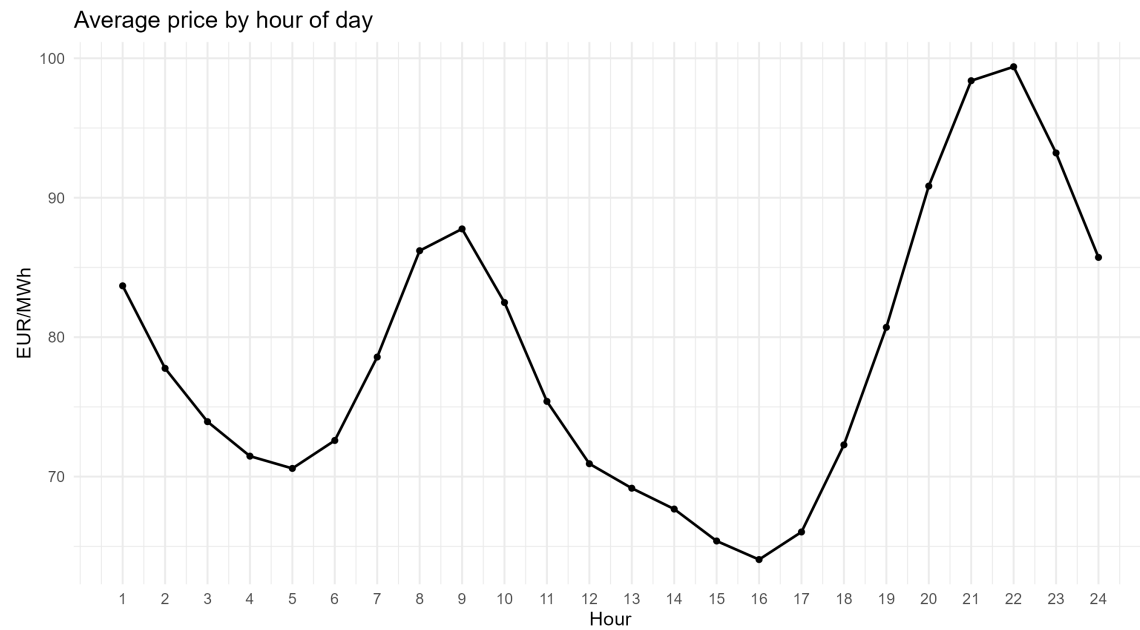


Figure 9: Average price by hour of day.

Hourly cleared-step profile

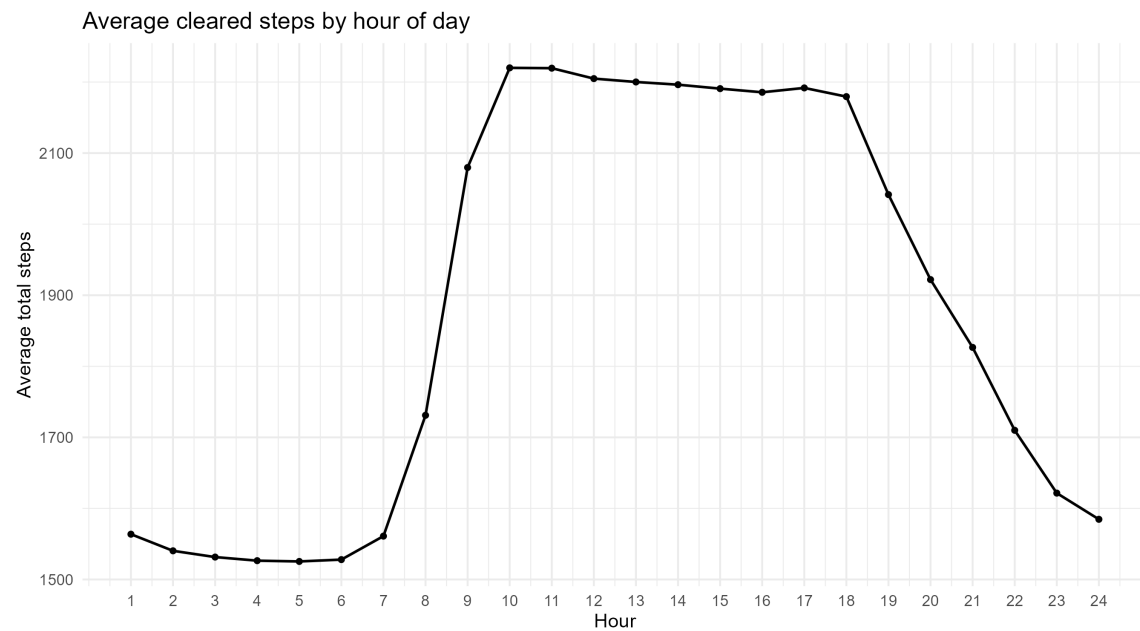


Figure 10: Average cleared steps by hour of day.

Monthly stress seasonality

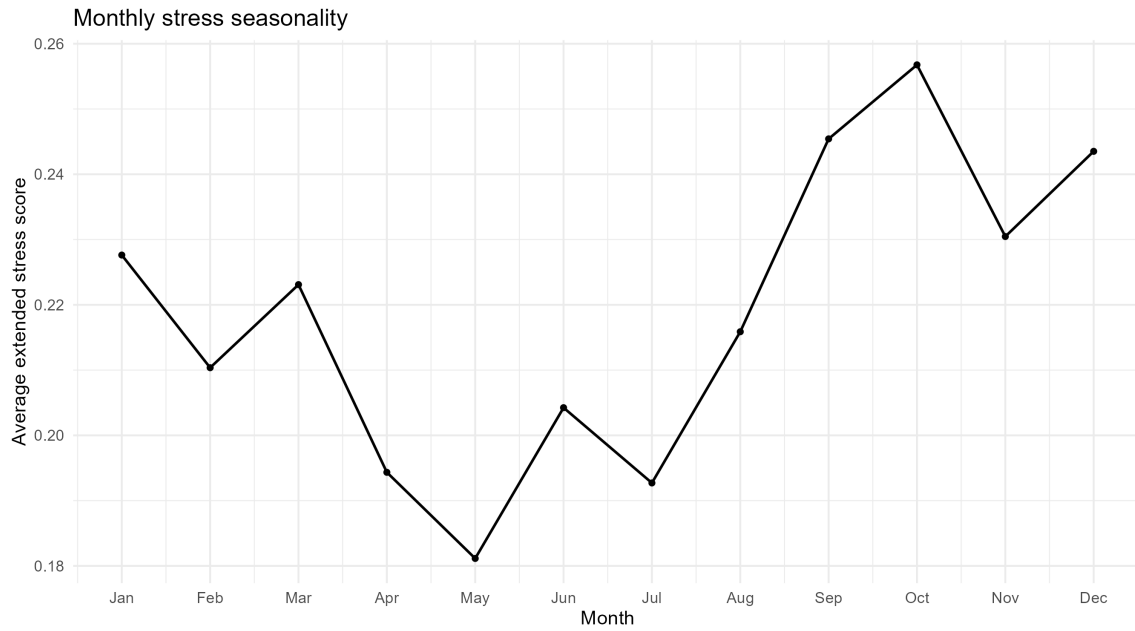


Figure 11: Monthly stress seasonality.